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AMD Graphics Core Next

Everything you wanted to know about AMD's next generation GPGPU architecture

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Graphics cards these days have various functions apart from the prime responsibility of improving the frame rates and realism in a game. With a lot of tasks requiring heavy parallel processing, the processor may not always be able to keep up with certain applications, or if it does, it wouldn't have enough resources to run other programs concurrently, as smoothly. This brings us to the concept of GPU Computing, wherein a graphics card not only helps improve your system's overall graphics related performance, but also enables speeding up certain computing tasks such as say rendering, transcoding, etc.

With its current generation of Radeon HD cards, AMD has introduced a new architecture called Graphics Core Next which will be seen in the HD 7900 series cards very soon.

With the HD 7000 series of cards, you have innovations such as a 28-nm manufacturing process, support for DX11.1 and support for C++. So, without further ado, let's have a look at the new architecture and try to figure out the advantages over the older gen VLIW architecture.

SIMD Out, CU In

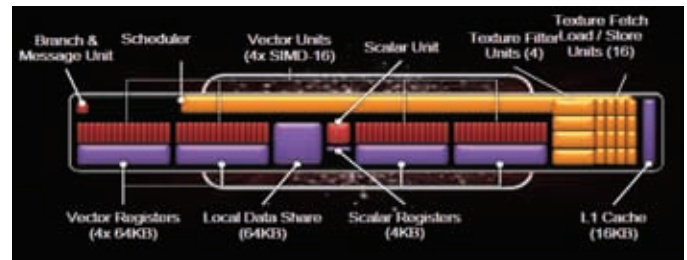
In the previous generation of cards, the basic building block – the Stream Processor – of the GPU was based on the Very Long Instruction Word 4 architecture. That means each stream processor has four math units (ALU) also known as Radeon cores, which execute instructions in parallel. Sixteen such stream processors make one VLIW 4 SIMD engine. Each

SIMD engine has 64 ALUs. This architecture is great for executing multiple operations from a similar task by breaking it up into small segments called Wavefronts. The compiler which does all the scheduling is optimised for graphic-intensive tasks involving mapping dot product math. In a game for instance, the compiler does the scheduling well in advance and it's static, that is it can't be changed on the fly.

In some scalar instructions seen in more general purpose applications, one wave front can't execute until another wavefront has been resolved. In such circumstances, the VLIW architecture seems lacking as the compiler can't change the wavefront queue as it has already been scheduled. This issue is known as a dependency which is common with general purpose compute related tasks.

In the diagram below, you have the wavefronts going from 1 to 10, with wavefronts 3, 6 and 7 having dependencies. With the VLIW4 architecture it will take four clock cycles to resolve all the wavefronts.

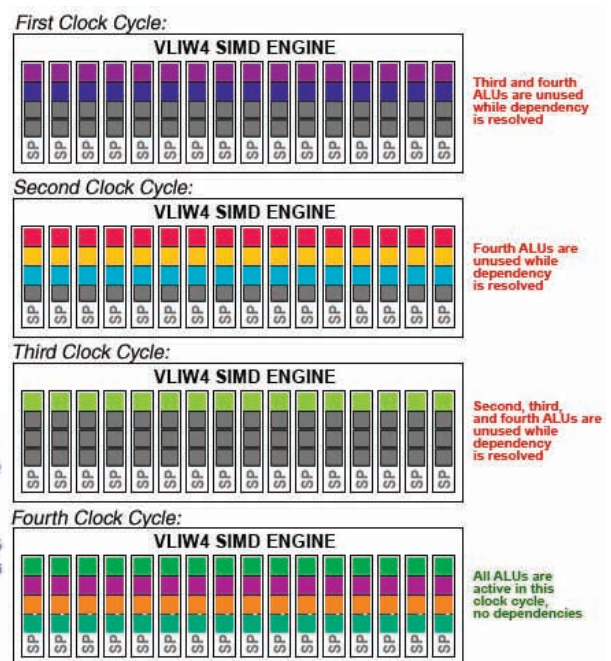
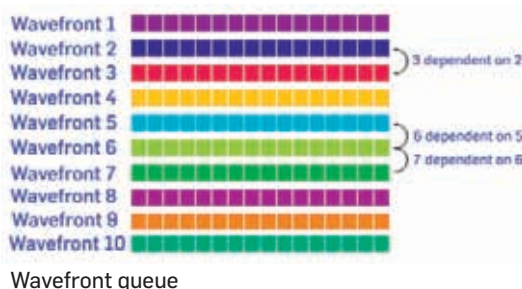
This issue is resolved with the help of a Compute Unit (CU), which is at the



The Compute Unit

heart of the Graphics Core Next. Just like a VLIW4 SIMD engine, the Compute Unit also has 64 ALUs, but they're grouped into four Vector Units – each having 16 ALUs. In a VLIW4 SIMD engine all the 16 stream processors were scheduled simultaneously by a compiler, but in the compute unit, each of the four vector units can be scheduled individually. There is a dedicated hardware scheduler – Branch Unit and MSG.

The immediate advantage is it takes care of dependencies. As seen on the next page, the 10 wavefronts are resolved in lesser time, as the hardware scheduler schedules



Dependency handling in VLIW4

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